Marco Notaro

Curriculum Vitae

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Education

Oct 2015-Feb 2019 **Ph.D. in Computer Science**, *Thesis Title: "Hierarchical Ensemble Methods for Ontology-based Predictions in Computational Biology"*, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re.

Oct 2012-Feb 2015 **M.Sc. in Molecular Biotechnology and Bioinformatics (Curriculum Bioinformatics)**, *Thesis Title: "Hierarchical Ensemble Methods for Biological Ontologies Structured as Directed Acyclic Graphs"*, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re – Final Grade: 110/110.

Oct 2009-Oct 2012 **B.Sc. in Industrial and Environmental Biotechnology (Curriculum Bioinformatics)**, *Thesis Title: "High Resolution Maps of Methylation and Acetylation of Histones in the Human Genome"*, Milan University, Bioscience Department, Advisor: Prof. Giulio Pavesi – Final Grade: 101/110.

Research and Work Experiences

Feb 2019-Present **Collaborator**, in the project "Data Model of Next-Generation DataBase Management System (DBMS)", Milan University, Computer Science Department. Supervisor: Prof. Marco Mesiti.

Oct 2015-Feb 2019 **PhD Research Activity**, at Computational Biology and Bioinformatics Group, Milan University, Computer Science Department, under the supervision of Prof. Giorgio Valentini. Some relevant research experiences alongside my PhD: • I was co-advisor of a master thesis in Computer Science; • I did Bioinformatics lectures for M.Sc. courses in Computer Science (link).

Mar 2018-May 2018 **Visiting PhD Student**, at CITIC-UGR, Centro de Investigación en Tecnologías de la Información y de las Cominucaciones, Granada University, Spain, under the supervision of Dr. Carlos Cano Gutierrez.

Sept 2016-Jan 2017 **CAFA3 Challenge**, international challenge for the prediction of protein function and for the prediction of genes associated with abnormal phenotypes.

June 2016-Aug 2016 **Visiting PhD Student**, at Charité - Universitätsmedizin Berlin, Institute for Human and Medical Genetics, joining in the *Computational Biology Group* under the supervision of Prof. Peter Robinson.

Oct 2015-Jan 2016 **Assistant**, for the course "Statistics and Data Analysis" (R Laboratory), Milan University, Computer Science Department, Advisor: Prof. Dario Malchiodi (link).

Mar 2015-May 2015 **Assistant**, for the course "Data Analysis" (R Laboratory), Milan University, Computer Science Department, Advisor: Anna Mari Zanaboni (link).

Feb 2014-Feb 2015 **Master Degree Internship**, at Computational Biology and Bioinformatics Group, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re.

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Summer Schools and International Meeting

12-16 Sept 2016 **Big Data for Precision Medicine and The 4th i2b2 European Meeting**, Pavia, IT.

Speaker at International Conferences

7-9 Sept 2016 **CIBB2017**, Cagliari, IT, Took part as speaker by presenting the following works 1. Ensembling Descendant Term Classifiers to Improve Gene - Abnormal Phenotype Predictions; 2. Disease Genes must Guide Data Source Integration in the Gene Prioritization Process.

Language Skills and Certifications

Italian Mother Tongue.

English Upper Intermediate Level, (Certified by Shenker Method -2013-).

Computational Skills

Main Programming Languages • **R**: machine learning; statistics and data analysis; data manipulation and data visualization (ggplot2) • **Perl**: parsing and analysis of raw text files; pattern matching by regular expression • **Bash**: scheduling computational experiments and executing commands in Linux environment

Side Programming Languages **JavaScript**: misc. and basic knowledge for maintaining my website • **Python**: misc. and basic knowledge for scientific usage (as numpy, scipy) • **MATLAB and C++**: some basic and misc. knowledge

Markup Languages • **HTML** and **CSS**: for maintaining my website • **LaTex**: for writing scientific papers and reports

Software Libraries • **HEMDAG** an R package available both on biconda and cran repository alongside a step-by-step tutorial (link); • **HPOparser/GOparser** (perl modules) (note: a more general perl module, named **OBOparser**, will be submitted to *cpan* shortly) For more details about what these packages do, please have a look at my Software Page

Operating systems • Linux (Ubuntu); • Windows

Research Interests

My research interests touch the fields of:

- Bioinformatics
- Computational Biology
- Machine Learning

To know further details about my research interests, please get a glimpse of my Research Page.

Awards

Prediction of Human Phenotype Ontology terms by means of Hierarchical Ensemble Methods: awarded by the International Medical Informatics Association (IMIA) as one of the five best "Knowledge Representation and Management" (KRM) papers of 2017 in the field of Medical Informatics (link)

Publications

Note: an updated list of all my papers can be also found on my Publications Page

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International Peer-Reviewed Journals

5. **Notaro M**, Frasca M, Petrini A, Valentini G. HEMDAG: a scalable and flexible hierarchical ensemble-based tool to improve Gene Ontology term predictions; 2019. Manuscript in preparation.

4. Zhou N, Jiang Y, . . . , **Notaro M**, . . . , Radivojac P, Friedberg I. The CAFA challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens. bioRxiv. 2019. Cold Spring Harbor Laboratory. doi:10.1101/653105. Available from: https://www.biorxiv.org/content/early/2019/05/29/653105.

3. Perlasca P, Frasca M, Ba CT, **Notaro M**, Petrini A, Casiraghi E, Grossi G, Gliozzo J, Valentini G, Mesiti M. UNIPred-Web: a web tool for the integration and visualization of biomolecular networks for protein function prediction. BMC Bioinformatics. 2019. 20(1):422:1–422:19. doi:10.1186/s12859-019-2959-2. Available from: https://doi.org/10.1186/s12859-019-2959-2.

2. Frasca M, Grossi G, Gliozzo J, Mesiti M, **Notaro M**, Perlasca P, Petrini A, Valentini G. A GPU-based algorithm for fast node label learning in large and unbalanced biomolecular networks. BMC Bioinformatics. 2018. 19-S(10):269–280. doi:10.1186/s12859-018-2301-4. Available from: https://doi.org/10.1186/s12859-018-2301-4.

1. **Notaro M**, Schubach M, Robinson PN, Valentini G. Prediction of Human Phenotype Ontology terms by means of hierarchical ensemble methods. BMC Bioinformatics. 2017. 18(1):449:1–449:18. doi:10.1186/s12859-017-1854-y. Available from: https://doi.org/10.1186/s12859-017-1854-y.

Proceedings of International Conferences and Peer-Reviewed Book Chapters

4. **Notaro M**, Schubach M, Frasca M, Mesiti M, Robinson PN, Valentini G. Ensembling Descendant Term Classifiers to Improve Gene - Abnormal Phenotype Predictions. In: Bartoletti M, Barla A, Bracciali A, Klau GW, Peterson L, Policriti A, Tagliaferri R, editors. Computational Intelligence Methods for Bioinformatics and Biostatistics. Cham: Springer International Publishing; 2019. doi:10.1007/978-3-030-14160-8_8. Available from: https://doi.org/10.1007/978-3-030-14160-8_8.

3. Frasca M, Fontaine JF, Valentini G, Mesiti M, **Notaro M**, Malchiodi D, Andrade-Navarro MA. Disease–Genes Must Guide Data Source Integration in the Gene Prioritization Process. In: Bartoletti M, Barla A, Bracciali A, Klau GW, Peterson L, Policriti A, Tagliaferri R, editors. Computational Intelligence Methods for Bioinformatics and Biostatistics. Cham: Springer International Publishing; 2019. doi:10.1007/978-3-030-14160-8_7. Available from: https://doi.org/10.1007/978-3-030-14160-8_7.

2. Robinson PN, Frasca M, Köhler S, **Notaro M**, Re M, Valentini G. A Hierarchical Ensemble Method for DAG-Structured Taxonomies. In: Schwenker F, Roli F, Kittler J, editors. Multiple Classifier Systems. Cham: Springer International Publishing; 2015. doi:10.1007/978-3-319-20248-8_2. Available from: https://doi.org/10.1007/978-3-319-20248-8_2.

1. Valentini G, Köhler S, Re M, **Notaro M**, Robinson PN. Prediction of Human Gene - Phenotype Associations by Exploiting the Hierarchical Structure of the Human Phenotype Ontology. In: Ortuño F, Rojas I, editors. Bioinformatics and Biomedical Engineering. Cham: Springer International Publishing; 2015. doi:10.1007/978-3-319-16483-0_7. Available from: https://doi.org/10.1007/978-3-319-16483-0_7.

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